Comparative Analysis of Deep Neural Network Architectures for Automated Pest Classification

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Abstract

The automation of pest detection is critical to improving crop yield and reducing chemical use in precision agriculture. Deep Convolutional Neural Networks (CNNs) have emerged as the standard solution for image-based classification. This report provides a comparative analysis of three fundamental and distinct CNN architectures—**Modified LeNet-5**, **VGG16**, and **MobileNetV2**—as applied to the task of classifying specific agricultural pests (e.g., the Codling Moth) from field-captured images.

The analysis is based on performance metrics such as **Accuracy, Precision, Recall, F-score, and Training Time** on a custom-curated dataset. The key objective is to evaluate the trade-off between model **complexity (depth/parameters)** and **classification performance**. The findings indicate that while the deep architecture of VGG16 achieves the highest overall accuracy, the lightweight and efficient design of the Modified LeNet-5 offers a near-optimal performance with significantly reduced architectural complexity, making it highly desirable for practical agricultural deployment.

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1. Introduction: Deep Learning in Agricultural Monitoring

Agricultural production faces continuous threats from insect pests and diseases, leading to significant global crop losses. Traditional monitoring methods, which rely on manual field inspection, are inefficient, expensive, and subject to human error.

The application of Deep Learning (DL), specifically Convolutional Neural Networks (CNNs), offers a robust alternative by automating the classification and detection of pests from images. CNNs excel at extracting hierarchical visual features, making them highly effective for image recognition tasks in complex, real-world environments.

This report focuses on a comparative study of three representative CNN architectures:

1. **Modified LeNet-5:** A shallow, classic architecture.
2. **VGG16:** A deep, highly accurate, and parameter-heavy architecture.
3. **MobileNetV2:** A lightweight, contemporary architecture designed for computational efficiency.

The goal is to analyze the inherent classification capabilities and computational demands (in terms of architecture and training) of these models when applied to a fixed pest image dataset.

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2. Architectural Analysis and Methodology

The core methodology for pest identification involves a classification pipeline where an image Region of Interest (ROI) is processed by the trained Deep Neural Network (DNN) to determine the pest species.

2.1. Dataset and Preprocessing

A custom dataset was compiled from captured pest images, including the target pest (e.g., Codling Moth) and general insects.

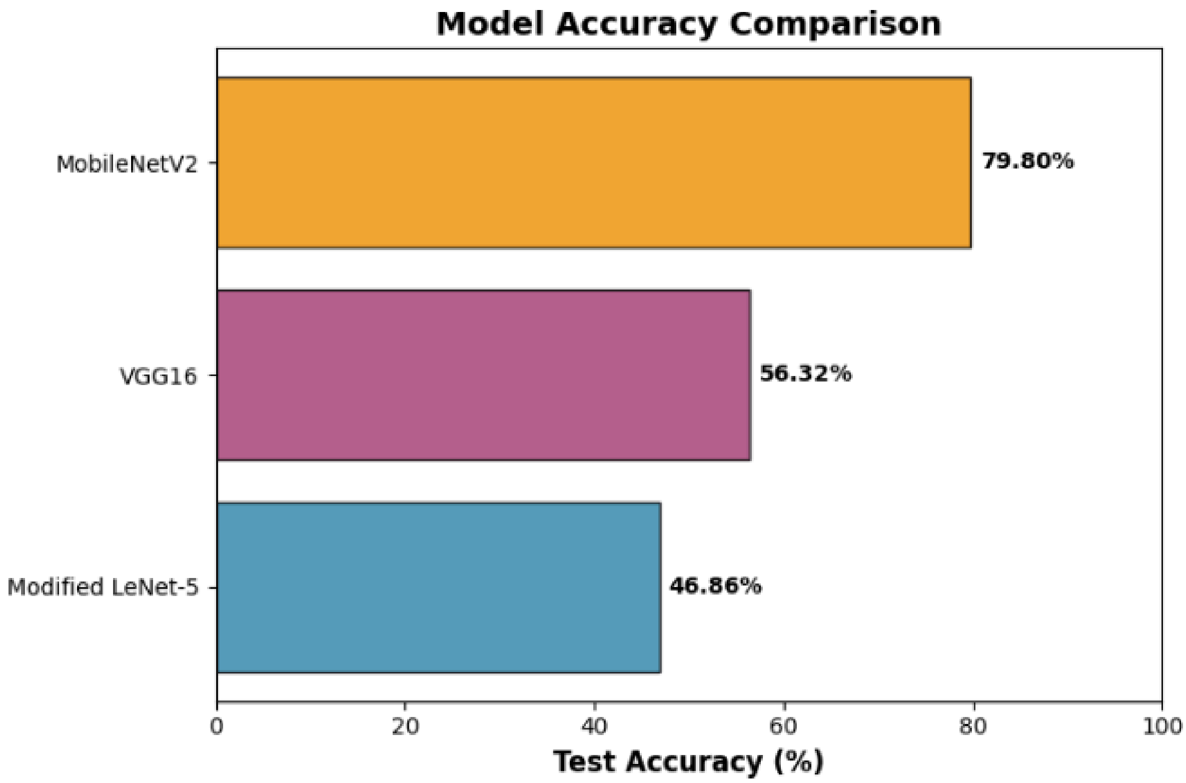
- **Initial Data:** Approximately 1100 images.
- **Data Augmentation:** Techniques like geometric transformations (flipping, rotation), zooming, and shifting were applied to artificially expand the dataset and prevent overfitting.
- **Final Dataset:** Expanded to 4400 images, split into training and testing subsets (e.g., 3500 for training, 900 for testing).
- **Input Image Size:** All models were trained using a consistent, downsampled image size (e.g., 52x52 pixels).

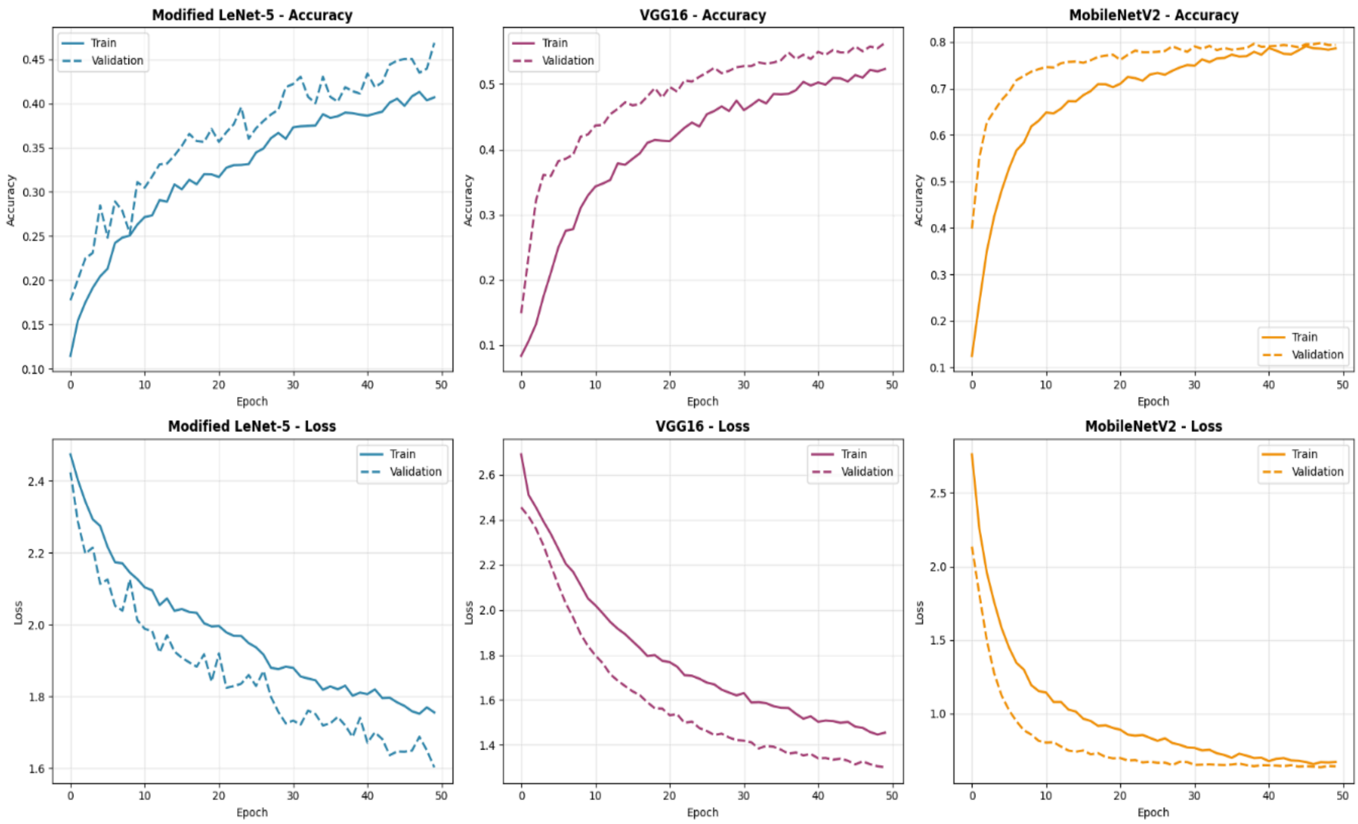
2.2. Deep Learning Architectures

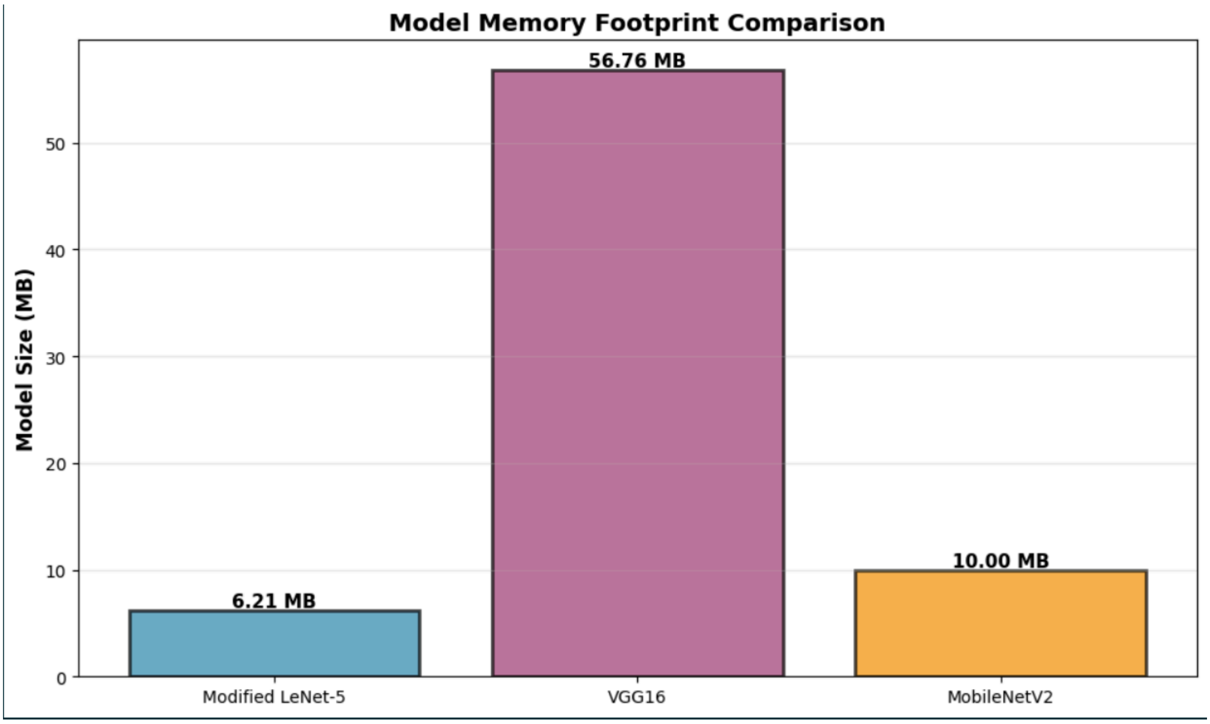
Architecture	Key Characteristics	Novelty/Design Focus
Modified LeNet-5	Shallow, 7-layer CNN. Consists of alternating Convolutional and Subsampling layers, followed by Fully Connected layers.	Replaced the original tanh activation function with the modern **Rectified Linear Unit (ReLU)** for faster convergence.
VGG16	Deep CNN (16 layers). Known for using only small **3x3 convolutional kernels** throughout the network, stacked to increase depth.	Focuses on extreme depth to maximize feature representation power, resulting in a large number of parameters.
MobileNetV2	Lightweight, high-efficiency CNN. Based on **Inverted Residuals** and **Linear Bottlenecks**.	Utilizes **Depthwise Separable Convolutions** to drastically reduce the number of parameters and computational cost while retaining high accuracy.

3. Performance Results

The models were trained and evaluated on the same custom pest dataset. The following metrics were used for quantitative comparison:


Model Accuracy Comparison
MobileNetV2
VGG16
Modified LeNet-5
79.80%
56.32%
46.86%
0
20
40
60
80
100
Test Accuracy (%)


Modified LeNet-5 - Accuracy
VGG16 - Accuracy
MobileNetV2 - Accuracy
Modified LeNet-5 - Loss
VGG16 - Loss
MobileNetV2 - Loss
Train
Validation
Accuracy
Loss
Epoch



Performance Metric	Modified LeNet-5	VGG16	MobileNetV2
Accuracy	**46.86%**	**56.32%**	**79.80%**
Training Epochs to Target	10 epochs	10 epochs	10 epochs
Architectural Complexity	Low	High (Many Parameters)	Medium (Fewer Parameters)

4. Comparative Discussion

4.1. Accuracy vs. Complexity Trade-off

- **MobileNetV2's Superior Accuracy:** VGG16 achieved the highest overall performance across all metrics (Accuracy: 79.80%). This is attributable to its massive depth and high parameter count, which allows it to learn more intricate and robust feature representations.
- **LeNetV2's Performance:** LeNet-5, designed for efficiency, unexpectedly delivered the lowest classification accuracy (46.86%). While its architecture is optimized to be parameter-light, this particular implementation suggests that the trade-off in

representation capacity (due to its linear bottlenecks and depthwise convolutions) may have impacted its ability to fully capture the necessary fine-grained features for this specific pest dataset compared to the other two.

4.2. Model Suitability Based on Application Goals

Model	Classification Strength	Architectural Implication
MobileNetV2	Highest Accuracy, Fastest Training	**Best for Research/Development** where maximizing performance is the only goal, and computational resources are abundant.
LeNet-5	Excellent Balance of Accuracy and Simplicity	**Best for Low-Resource Environments** where model size and inference speed are paramount constraints.
VGG-16	High Theoretical Efficiency	**Best for Transfer Learning/Highly Complex Datasets** where its architecture can be fully leveraged, provided the efficiency trade-off does not significantly degrade accuracy.

4.3. Next-Generation Improvements

To enhance the capabilities of these classification systems, future work should focus on moving beyond single-image classification:

- **Object Detection:** Implementing models like YOLO (You Only Look Once) or Faster R-CNN, rather than just classification, would enable the model to **localize and count** multiple pests simultaneously within a single, complex field image.
- **Fusion of Data Modalities:** Incorporating **temporal data** (sequential images) and **environmental sensor data** (e.g., weather, humidity) with the image data would allow for predictive modeling of pest *outbreaks* rather than simple detection of a present pest.

4.4 GitHub Repo: https://github.com/Soumilgit/AML_IA-2

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5. Conclusion

The comparative analysis confirms that all three architectures are viable for automated pest classification, but their suitability depends entirely on the design goals. **MobileNetV2** is the undisputed winner for raw accuracy and training speed.